

PBR-H95 Adaptive Mantissa Quantization

PoC and Evaluation Guide

A quality-constrained hybrid successor to the lossless PBR experiments

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Research status: The earlier lossless RAW/Huffman/Context and matrix V1–V3 experiments correctly reached the approximately seven-bit mantissa wall on real Qwen. PBR-H95 changes the objective: minimize serialized model size while preserving at least 95% of measured BF16 model quality. This is a proposed PoC plan, not a completed Qwen result.

1. Executive Summary

Accepting a 95% model-quality floor changes the work from lossless compression to calibrated quantization. PBR-H95 will retain a selectable number of mantissa bits per tensor or tensor group, reconstruct a deterministic quantized BF16 model, and then apply exact storage coding to that quantized representation. The storage codec must reproduce the quantized checkpoint exactly, while the quantized checkpoint must retain at least 95% of the original BF16 evaluation score.

Primary objective: Minimize complete serialized bits per weight subject to model-quality retention $\geq 95\%$.

```
minimize: complete_serialized_BPW
subject to: score_quantized / score_BF16 >= 0.95
```

2. Define “95% Accuracy” Correctly

The acceptance target must describe model behavior, not merely weight similarity.

Possible definition	Use?	Reason
95% identical BF16 words	No	A small number of changed critical weights can cause disproportionate quality loss.
95% numerically close weights	Diagnostic only	Weight-space similarity does not guarantee output quality.
$\geq 95\%$ of baseline model score	Yes	Directly measures the intended model behavior on a fixed evaluation set.

Recommended quality-retention equation:

```
quality_retention = compressed_model_score / original_BF16_score
pass = quality_retention >= 0.95
```

Use a fixed, versioned evaluation set and report the metric appropriate to the model, such as perplexity, task accuracy, exact match, pass@k, domain golden-set score, groundedness, and refusal/safety behavior.

3. Rate Opportunity

Using the current approximate field rates, the complete model rate can be estimated as:

```
total_BPW ≈ sign_BPW + exponent_BPW + retained_mantissa_BPW + metadata_BPW
total_BPW ≈ 1.00 + 2.62 + retained_mantissa_bits + metadata
```

Retained mantissa bits	Approx. total before new metadata	Interpretation
7	10.62 BPW	Current lossless region
6	9.62 BPW	Conservative lossy target
5	8.62 BPW	Likely first useful operating point

4	7.62 BPW	Potential ≤ 8 -BPW region
3	6.62 BPW	Aggressive; requires strong calibration
2	5.62 BPW	Very aggressive
1	4.62 BPW	High quality risk
0	3.62 BPW	No mantissa information; generally unrealistic

A complete 8-BPW target becomes plausible if the weighted average retained mantissa budget is near four bits and metadata remains small. A 4-BPW target is not realistic with the present independent sign and exponent rates because they already consume approximately 3.62 BPW.

4. PBR-H95 Candidate Modes

Each tensor or tensor group may select one of the following representations:

- EXACT_BF16 — original seven-bit mantissa retained.
- MANTISSA_6 — six retained mantissa bits with deterministic rounding.
- MANTISSA_5 — five retained mantissa bits.
- MANTISSA_4 — four retained mantissa bits.
- MANTISSA_3 — three retained mantissa bits for robust tensors only.
- Optional later candidates — INT8, INT6, INT4, or external calibrated quantizers for comparison.

The controller must optimize size and quality jointly. It must not choose the smallest representation if that choice violates the global quality floor.

5. Mantissa Rounding Method

For a seven-bit mantissa, retain the most significant k bits and round the removed portion to nearest. Avoid simple truncation because it creates a consistent downward bias.

```
removed_bits = 7 - keep_bits
rounded = mantissa + (1 << (removed_bits - 1))
quantized = rounded >> removed_bits
reconstructed_mantissa = quantized << removed_bits
```

Important: rounding can overflow the mantissa and require an exponent increment. Therefore the final implementation must quantize the full BF16 word with explicit normalization, special-value handling, and saturation rules. It must preserve NaN/Inf policy, signed zero policy, and deterministic behavior.

6. Adaptive Precision Policy

Start conservatively and use calibration to reduce precision only where the quality cost is low.

Tensor family	Initial precision policy	Rationale
Embeddings	7 bits initially	Token representation can be sensitive and widely reused.
LM/output head	7 bits initially	Direct influence on output logits.
Normalization weights	7 bits	Small tensors with disproportionate numerical impact.
Bias and small vectors	7 bits	Small storage contribution; little reason to risk quality.
Attention projections	Test 6 $\rightarrow$ 5 $\rightarrow$ 4 bits	Large tensors and meaningful storage opportunity.
MLP gate/up/down	Test 6 $\rightarrow$ 5 $\rightarrow$ 4 $\rightarrow$ 3 bits	Large share of weights; likely primary compression target.
First/last blocks	Higher precision initially	Boundary layers may be more sensitive.

Outlier channels	Restore 6–7 bits	Protect large or activation-salient weights.
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7. Calibration and Search Algorithm

Use a fixed calibration set and a separate held-out evaluation set. Never optimize and report on the same examples.

1. Measure and save the original BF16 baseline score.
2. Generate candidate precision changes for one tensor or layer group at a time.
3. Evaluate calibration degradation and serialized bytes saved.
4. Compute utility: bytes saved divided by quality loss.
5. Apply the highest-utility changes first.
6. Re-evaluate the combined model periodically because errors interact.
7. Stop before the held-out quality-retention score falls below 95%.
8. Restore precision to sensitive tensors until all global and subgroup gates pass.

```
utility = bytes_saved / max(calibration_quality_loss, epsilon)
accept candidate only if projected_global_quality >= 0.95
```

8. Quality-Loss Budget

Treat the permitted 5% degradation as a global budget. Do not allow each layer to lose 5%, because layer errors can accumulate and interact.

Budget category	Illustrative starting allocation
Embeddings and output head	0.25 percentage points
Normalization and biases	0.10 percentage points
Attention blocks	1.75 percentage points
MLP blocks	2.50 percentage points
Interaction reserve	0.40 percentage points

These allocations are initialization heuristics, not final claims. Replace them with measured sensitivity data from the selected Qwen model and evaluation suite.

9. Reuse of Existing PBR Components

The earlier work remains useful after quantization. Quantization may create more repeated or skewed mantissa patterns, allowing the existing exact storage candidates to become profitable.

- RAW storage for quantized mantissas.
- Huffman coding for skewed quantized symbols.
- Context coding for predictable quantized streams.
- Bit-packed residual-pair dictionaries for repeated quantized residual pairs.
- Tensor-level ALL_* and MIXED selection after complete metadata accounting.

The complete pipeline is:

```
BF16 checkpoint
-> sensitivity analysis
-> per-tensor precision selection
-> deterministic mantissa quantization
-> exact entropy coding of quantized fields
-> serialized hybrid container
-> exact restoration of quantized checkpoint
-> model-quality evaluation
```

10. Two Separate Acceptance Gates

10.1 Storage/container correctness

- Quantized reference words equal decoded quantized words exactly.
- Zero differing words relative to the quantized reference.
- Matching SHA-256 for quantized reference and decoded checkpoint.
- Deterministic output bytes for identical input and configuration.
- Reported serialized size equals physical file size.
- Malformed headers, payloads, modes, and checksums are rejected.

10.2 Model-quality acceptance

- Held-out aggregate score retention is at least 95%.
- No critical task or subgroup falls below its separately defined floor.
- No NaNs, infinities, or unstable generation are introduced.
- Domain golden-set behavior remains acceptable.
- Safety and refusal behavior are evaluated separately.
- Latency, throughput, and memory are measured on the target runtime.

11. PoC Phases

Phase	Scope	Output	Gate
H95-A	Offline BF16 rounding tests	Per-tensor numeric-error report	Exact deterministic quantized words
H95-B	Layer/tensor sensitivity sweep	Bytes-saved vs quality-loss ranking	Stable calibration behavior
H95-C	Greedy global precision allocation	Quality/size Pareto frontier	≥95% held-out retention
H95-D	Exact PBR container for selected model	Encoded file and independent decoder	SHA-256 and size accounting pass
H95-E	Target runtime benchmark	Quality, memory, tok/s, p95 latency	Deployment criteria pass

12. Required Reports

- Model identifier and exact source revision.
- Calibration and held-out dataset identifiers.
- Original BF16 score and quantized score.
- Aggregate and subgroup quality retention.
- Precision selected for every tensor.
- Counts and weights at 7/6/5/4/3 retained bits.
- Sign, exponent, mantissa, metadata, and total BPW.
- Physical encoded file size and compression ratio.
- Original, quantized-reference, and restored SHA-256 values.
- Differing words between quantized reference and restored model.
- Peak memory, throughput, latency, and runtime compatibility.

13. Recommended Target Ladder

Quality floor	Desired result	Decision
100%	Lossless reference near 10.6 BPW	Baseline
99%	Measure first meaningful lossy gain	Conservative deployment candidate

98%	Seek lower average precision	Balanced candidate
97%	Approach approximately 8 BPW	Evaluate domain risk carefully
95%	Minimum accepted quality floor	Stop quantizing when reached

14. Go/No-Go Criteria

GO

- Complete model rate is materially below the 10.6-BPW lossless result.
- Held-out model-quality retention is at least 95%.
- Critical subgroups remain above their individual quality floors.
- The quantized checkpoint round-trips exactly through the storage container.
- The target runtime demonstrates acceptable latency and memory behavior.

NO-GO

- The result reports weight similarity without model-level evaluation.
- Calibration score passes but held-out score fails.
- Aggregate quality passes while a critical subgroup collapses.
- Container decoding differs from the quantized reference.
- Metadata or runtime conversion removes the apparent storage advantage.
- The ≤ 8 -BPW target requires falling below the 95% quality floor.

15. Recommended First Experiment

Begin with per-tensor precision because metadata is negligible across approximately 290 tensors and the implementation is easier to validate than per-channel coding.

9. Keep embeddings, LM head, normalization, biases, first block, and last block at seven mantissa bits.
10. Set large attention and MLP matrices to five retained mantissa bits.
11. Evaluate calibration and held-out quality.
12. Reduce low-sensitivity MLP tensors from five to four bits.
13. Reduce only clearly robust tensors to three bits.
14. Restore precision to any tensor group that causes disproportionate loss.
15. Build the Pareto curve at 100%, 99%, 98%, 97%, and 95% quality.
16. After the precision map is frozen, apply exact PBR entropy coding and report physical file size.

Recommended research direction: PBR-H95 per-tensor adaptive mantissa precision is a more credible path to approximately 8 total BPW than another lossless predictor sweep. The quantized model may be lossy relative to BF16, but its serialized representation must remain exact, deterministic, independently decodable, and fully accounted.